

Def. Let  $V$  be a vector space.

Given a vector  $\alpha \in V$  and linear operator  $T: V \rightarrow V$ , the subspace

$$Z(\alpha, T) \stackrel{\text{def}}{=} \{g(T)\alpha \mid g(t) \in F[t]\} = \text{span}\{T^k \alpha \mid k \in \mathbb{N} \cup \{0\}\}.$$

is called the  $T$ -cyclic subspace generated by  $\alpha$

Def. Let  $V$  be a vector space. Given  $\alpha \in V$  and a linear operator  $T$  on  $V$ , an ideal

$$M(\alpha, T) = \{g(t) \in F[t] \mid g(T)\alpha = 0\}$$

is called the  $T$ -annihilator of  $\alpha$ , and the unique monic generator is also called  $\mu$ .

Thm. Let  $\alpha$  be any non-zero vector in  $V$  and let  $p_\alpha$  be the  $T$ -annihilator of  $\alpha$ .

1)  $\deg p_\alpha = \dim Z(\alpha, T)$

2)  $Z(\alpha, T) = \text{span}\{T^k \alpha \mid 0 \leq k \leq \deg p_\alpha\}$

3) The minimal polynomial of  $T|_{Z(\alpha, T)}$  is  $p_\alpha$ .

Proof. Given a polynomial  $g \in F[x]$ , the Euclidean Algorithm gives

$$g = p_\alpha \cdot q + r \quad \text{for some } q, r \in F[x], \text{ with } r = 0 \text{ or } \deg r < \deg p_\alpha.$$

Then,  $g(T)\alpha = p_\alpha(T)g(T)\alpha + r(T)\alpha = r(T)\alpha$ . Since either  $r = 0$  or  $\deg r < \deg p_\alpha$ , the vector  $r(T)\alpha$  is a linear combination of  $\{\alpha, T\alpha, T^2\alpha, \dots, T^{\deg p_\alpha - 1}\alpha\}$ .

Since  $g$  was arbitrary,  $Z(\alpha, T)$  is spanned by  $\{T^n \alpha \mid n = 0, 1, \dots, \deg p_\alpha - 1\}$ .

Moreover, these are linearly independent, because: if there exists non-trivial representation

$$c_0 \alpha + c_1 T\alpha + \dots + c_{\deg p_\alpha - 1} T^{\deg p_\alpha - 1} \alpha = 0,$$

then it contradicts to the minimality of  $p_\alpha$ . It proves 1) and 2).

To show the third assertion, let  $U = T|_{Z(\alpha, T)}: Z(\alpha, T) \rightarrow Z(\alpha, T): \alpha \mapsto T\alpha$ .

It is well-defined, since the cyclic subspace is invariant under  $T$ .

Given a polynomial  $g \in F[x]$ ,

$$p_\alpha(U)g(T)\alpha \underset{g(T)\alpha \in Z(\alpha, T)}{=} p_\alpha(T)g(T)\alpha = g(T)p_\alpha(T)\alpha = g(T)0 = 0.$$

$$g(T)\alpha \in Z(\alpha, T)$$

So,  $p_\alpha(U)$  is identically zero. Moreover,  $h(U)$  cannot be zero if  $\deg h < \deg p_\alpha$ , because  $h(U)\alpha = h(T)\alpha = 0$ , then it contradicts to the minimality.

### Companion Matrix.

Let  $W$  be a vector space of dimension  $k \in \mathbb{N}$ , and  $T: W \rightarrow W$  be a linear operator which have a cyclic vector  $\alpha \in W$ . By the Theorem above,  $\{\alpha, T\alpha, \dots, T^{k-1}\alpha\}$  forms a basis for  $W$ . As above fact and the Cayley-Hamilton Theorem, let

$$p_\alpha(x) = x^k + c_{k-1}x^{k-1} + \dots + c_1x + c_0$$

be the minimal polynomial (and also characteristic) for  $T$ .

Define  $\beta = \{T^{n-1}\alpha \mid n=1, 2, \dots, k\}$ , denote  $d_n = T^{n-1}\alpha$ . Then,

$$0 = 0 \cdot \alpha = p_\alpha(T)\alpha = T^k\alpha + c_{k-1}T^{k-1}\alpha + \dots + c_1T\alpha + c_0\alpha$$

$$\Rightarrow T^k\alpha = -c_0\alpha - c_1T\alpha - \dots - c_{k-1}T^{k-1}\alpha$$

$$= -c_0d_1 - c_1d_2 - \dots - c_{k-1}d_k$$

$$[T]_\beta = \begin{bmatrix} 0 & 0 & \dots & 0 & -c_0 \\ 1 & 0 & \dots & 0 & -c_1 \\ 0 & 1 & \dots & 0 & -c_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & \dots & 0 & 1 & -c_{k-1} \end{bmatrix} = \begin{bmatrix} 0 & \dots & 0 & -c_0 \\ \vdots & & & \vdots \\ I_{k-1} & & & \vdots \\ & & & -c_{k-1} \end{bmatrix}.$$

Directly, we obtain that: on a fin. dim space  $T$ ,

A linear operator  $T$  has a cyclic vector

if and only if

the minimal polynomial of  $T$  and characteristic polynomial of  $T$  are identical.

# Exercises in Hoffman, Sec 7.2

1. Every linear operator  $T$  on two-dimensional either:

$T$  has a cyclic vector or  $T$  is a scalar multiple of the identity.

Proof. Suppose that  $T$  has no cyclic vector. That is, for all  $\alpha \in V$ ,

$\{\alpha, T\alpha\}$  is linearly dependent. Therefore, there is  $c \in F$  such that  $c\alpha = T\alpha$ .

Let  $\alpha_1 \in V$  be a non-zero vector. For some  $c_1 \in F$ ,  $c_1\alpha_1 = T\alpha_1$ .

Now, let  $\alpha_2 \in V \setminus \langle \alpha_1 \rangle$ . Then, similarly,  $c_2\alpha_2 = T\alpha_2$ , for some  $c_2 \in F$ .

By the construction,  $\alpha_1 - \alpha_2 \neq 0$ , and there is  $c_3 \in F$  such that  $c_3(\alpha_1 - \alpha_2) = T(\alpha_1 - \alpha_2) = c_1\alpha_1 - c_2\alpha_2$ , so

$$(c_3 - c_1)\alpha_1 = (c_3 - c_2)\alpha_2$$

But, since  $\alpha_2 \in V \setminus \langle \alpha_1 \rangle$ ,  $c_3 - c_1 = c_3 - c_2 = 0$ , that is,  $c_1 = c_3 = c_2$ .

Now, since  $\{\alpha_1, \alpha_2\}$  is basis for  $V$ ,  $T = c_1 I$ .  $\square$

4. If  $T^2$  has a cyclic vector, then  $T$  has a cyclic vector.

Proof. Let  $\alpha \in V$  be the cyclic vector of  $T^2$ . That is,

$\mathcal{B} = \{T^{2k}\alpha \mid k=0,1,\dots\}$  spans the space  $V$ . Clearly,  $\mathcal{B}' = \{T^k\alpha \mid k=0,1,\dots\}$  satisfies  $\mathcal{B} \subseteq \mathcal{B}'$ , therefore  $V = \langle \mathcal{B} \rangle \subseteq \langle \mathcal{B}' \rangle = V$ .

But the converse does not hold: let  $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ ,  $\alpha = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ . Then,  $\{\alpha, A\alpha\} = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$  spans  $F^2$  but

$A^2 \equiv 0$  has no cyclic vector.

5. Let  $V$  be an  $n$ -dimensional over  $F$ , and let  $N$  be a nilpotent on  $V$ . Suppose that  $N^{n-1} \neq 0$  and let  $\alpha \in V$  be any vector such that  $N^{n-1}\alpha \neq 0$ . Prove that  $\alpha$  is a cyclic vector for  $N$ .

Proof Since  $N$  is nilpotent operator,  $N^k = 0$  for some  $k \geq 0$ , therefore the minimal polynomial of  $N$  must be power of  $x$ .

But, since  $N^{n-1} \neq 0$ , so  $N, N^2, \dots, N^{n-2}$  are not zero,

by the Cayley-Hamilton Theorem,  $N^n = 0$ .

For the given  $\alpha \in V$ ,  $N^{n-1}\alpha \neq 0$  implies that  $\alpha, N\alpha, N^2\alpha, \dots, N^{n-1}\alpha$  are all not zero. (if  $N^{n-k}\alpha = 0$ , then  $N^{k-1} \cdot N^{n-k}\alpha = 0$ ) Moreover, these are mutually distinct, because: for  $0 \leq i < j \leq n-1$ ,  $N^i\alpha = N^j\alpha$  implies  $N^{j+n-i}\alpha = N^{i+n-j}\alpha = 0$ ,

but  $i+n-j < n$ , it is contradiction. Moreover, these are linearly independent.

Suppose that  $C_0\alpha + C_1N\alpha + \dots + C_{n-1}N^{n-1}\alpha = 0$ ,  $C_i \in F$ .

Take  $N^{n-1}, N^{n-2}, \dots, N$  to the both side, then we obtain

$C_0 = 0, C_1 = 0, \dots, C_{n-1} = 0$ . So, the set  $\{\alpha, \dots, N^{n-1}\alpha\}$  is linearly independent set of  $n$ -elements in the  $n$ -dimensional space  $V$ , so proved.  $\square$

7. Let  $V$  be an  $n$ -dimensional, and  $T: V \rightarrow V$  is diagonalizable linear operator. Then,  $T$  has a cyclic vector if and only if  $T$  has  $n$ -distinct eigenvalues.

Proof. Suppose that  $T$  has a cyclic vector  $\alpha \in V$ .

Since  $T$  is diagonalizable, there are linearly independent  $n$ -vectors  $\alpha_1, \dots, \alpha_n$  such that  $\alpha = \alpha_1 + \dots + \alpha_n$  and  $T\alpha_i = C_i \alpha_i$  for some  $C_i \in F$ .

Observe that for each  $k \geq 0$ ,

$$T^k \alpha = T^k (\alpha_1 + \dots + \alpha_n) = \sum_{i=1}^n C_i^k \alpha_i.$$

These are linearly independent if and only if

$\begin{bmatrix} 1 & C_1 & \dots & C_1^{n-1} \\ \vdots & \vdots & & \vdots \\ 1 & C_n & \dots & C_n^{n-1} \end{bmatrix} x = 0$  has only trivial solution, so the matrix is invertible. It implies  $C_1, \dots, C_n$  are distinct.

Conversely, let  $\beta = \{\alpha_1, \dots, \alpha_n\}$  be a basis of eigenvectors for  $T$  corresponding to distinct scalar values  $C_1, \dots, C_n$ . Then,  $\alpha = \alpha_1 + \dots + \alpha_n$  is a cyclic vector, because it is enough to show that  $\{\alpha, T\alpha, \dots, T^{n-1}\alpha\}$  is linearly independent.

Suppose that

$$\begin{aligned} x_0 \alpha + x_1 T\alpha + \dots + x_{n-1} T^{n-1} \alpha &= (x_0 + x_1 C_1 + \dots + x_{n-1} C_1^{n-1}) \alpha_1 \\ &\quad + (x_0 + x_1 C_2 + \dots + x_{n-1} C_2^{n-1}) \alpha_2 \\ &\quad \vdots \\ &\quad + (x_0 + x_1 C_n + \dots + x_{n-1} C_n^{n-1}) \alpha_n = 0. \end{aligned}$$

Since  $\alpha_1, \dots, \alpha_n$  are linearly independent, therefore

$$x_0 + x_1 C_1 + \dots + x_{n-1} C_1^{n-1} = \dots = x_0 + x_1 C_n + \dots + x_{n-1} C_n^{n-1} = 0.$$

It equals to  $\begin{bmatrix} 1 & C_1 & \dots & C_1^{n-1} \\ \vdots & \vdots & & \vdots \\ 1 & C_n & \dots & C_n^{n-1} \end{bmatrix} \begin{bmatrix} x_0 \\ \vdots \\ x_{n-1} \end{bmatrix} = 0$ . The matrix is called Vandermonde, it is invertible when  $C_1, \dots, C_n$  are distinct. Therefore, it has only trivial solution,  $x_0 = \dots = x_{n-1} = 0$ . So  $\{\alpha, \dots, T^{n-1}\alpha\}$  is linearly independent.

8. Let  $T$  be a linear operator on the finite dimensional space  $V$ . If  $T$  is cyclic and  $T$  commute with a linear operator  $U$ , then  $U$  is a polynomial in  $T$ .

Proof. We want to show that: for some polynomial  $g(t) \in F[t]$ ,

$$U\beta = g(T)\beta \quad \text{for all } \beta \in V.$$

Let  $\alpha \in V$  be the cyclic vector of  $T$ . Then,  $\{\alpha, T\alpha, \dots, T^{n-1}\alpha\}$  forms basis for  $V$ . Therefore, for some scalars  $C_0, \dots, C_{n-1}$ ,

$$U\alpha = C_0\alpha + C_1T\alpha + \dots + C_{n-1}T^{n-1}\alpha.$$

Now, put  $g(t) = C_0 + C_1t + \dots + C_{n-1}t^{n-1}$ . Then

$$U\alpha = g(T)\alpha,$$

$$UT\alpha = T U\alpha = T g(T)\alpha = g(T)T\alpha$$

$$\vdots$$

$$UT^{n-1}\alpha = g(T)T^{n-1}\alpha$$

Thus,  $U = g(T)$ .  $\square$